

Theorem: Julia set $J(f)$ is a perfect set and is therefore uncountable.

Proof: Let $v \in J(f)$ and let U be any neighborhood of v . We have to show that U contains points of $J(f)$ other than v .

We consider three cases.

Case (i) Suppose v is not a fixed point or periodic point of f (i.e. $f^p(v) = v$ for some $p \in \mathbb{N}$).
 Since $J(f) = \bigcup_{k=1}^{\infty} \bar{f}^k(v)$, U contains a point w of $\bar{f}^k(v) \subseteq J(f)$ for some $k \in \mathbb{N}$.
 Since $f^k(v) \neq v$, $w \neq v$.

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Case (ii): $f(w) = v$.

If $f(z) = v$ has no solution other than v , by the proof of Lemma 14.7, $v \notin J(f)$, which is a contradiction.

$\therefore \exists w \neq v$ s.t. $f(w) = v$, $w \in J(f)$.

Then $\exists$ a point u of $\bar{f}^k(w)$ for some $k \in \mathbb{N}$.

$\nexists u \in \bar{f}^{-(k+1)}(v)$

Also, $u \neq v$ because $f^k(u) = w \neq v$.

Case (iii): $f^p(v) = v$ for some $p \geq 2$.

Since $J(f^p) = J(f)$, by applying case (ii) to f^p , we get a point $u \in J(f^p) = J(f)$ s.t. $u \in U$ & $u \neq v$.

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Defn: Let f be any polynomial.
 w is called a periodic point if
 $f^p(w) = w$ for some $p \in \mathbb{N}$.
 The least such p is called the period of w .
 $\{w, f(w), \dots, f^{p-1}(w)\}$ is called an orbit.
 Let w be a periodic point of period p
 and let $\lambda = (f^p)'(w)$.
 Then w is called an attracting periodic point if $|\lambda| < 1$, and w is called
 a repelling periodic point if $|\lambda| > 1$.
Example: $f(z) = z^2$. Find the periodic points
 of f and determine which ones are
 attracting periodic pts and which ones are
 repelling periodic points.

Theorem: Let f be any polynomial of degree ≥ 2 .
 Then the Julia set $J(f)$ is the closure of the
 repelling periodic points of f .

Proof: Let w be a repelling periodic point
 of f of period p . Then w is a repelling fixed
 point of $g = f^p$.

Claim: $w \in J(g)$.
 (Since $J(f^p) = J(f)$, this will imply
 that $w \in J(f)$).

For this it is enough to show that the
 family $\{g^k\}_{k=1}^{\infty}$ is not normal at w .

Suppose $\{g^k\}$ is normal at w .
 Then $\exists$ an open set U containing w such that
 a subsequence of $\{g^k\}$ converges uniformly to
 an analytic function g_0 ($\because g^k(w) = w \rightarrow \infty$)

Then $(g^k)'(z) \rightarrow g_0'(z) \quad \forall z \in U$.

But $(g^k)'(w) = (g'(w))^k$ (because $g(w)=w$ & by chain rule)

Since w is a repelling fixed point of g ,

$$|g'(w)| > 1.$$

$$\therefore |(g^k)'(w)| = |g'(w)|^k \rightarrow \infty \text{ as } k \rightarrow \infty,$$

which is a contradiction to $(g^k)'(w) \rightarrow g_0'(w)$.

Hence, every repelling periodic point of f

is in $J(f)$.

Since $J(f)$ is closed, the closure of the set of repelling periodic points of f is a subset of $J(f)$.

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Let $E = \{w \in J(f) \text{ such that } \exists v \neq w \text{ with } f(v)=w \text{ and } f'(v) \neq 0\}$.

Suppose $w \in E$.

Then $\exists$ an open set V containing w on which we can find a local analytic inverse

$$f^{-1}: V \rightarrow \mathbb{C} \setminus V$$

such that $f^{-1}(w) = v \neq w$. (by the inverse function theorem)

Define a family of analytic functions $\{h_k\}$

on V by

$$h_k(z) = \frac{f^k(z) - z}{f^{-1}(z) - z}$$

Let U be any open neighborhood of w with $U \subseteq V$.

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Since $w \in J(f)$, the family $\{f^k\}$ is not normal on U . This implies the family $\{h_k\}$ is not normal on U .

By the Montel's theorem, $h_k(z)$ must take either the value 0 or 1 for some $k \in \mathbb{N}$ and $z \in U$.
Now, $h_k(z) = 0 \Leftrightarrow f^k(z) = z \Leftrightarrow z$ is a periodic pt. of f .

Also, $h_k(z) = 1 \Leftrightarrow f^k(z) - z = \bar{f}^1(z) - z$
 $\Leftrightarrow f^k(z) = \bar{f}^1(z) \Leftrightarrow f^{k+1}(z) = z$
 $\Leftrightarrow z$ is a periodic pt. of f .

Thus, U contains a periodic point of f .
This implies that w is in the closure of the periodic points of f for all $w \in E$.

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By Fatou-Shishikura inequality, it is known that for a polynomial of degree $d \geq 2$, there are at most $(d-1)$ non-repelling periodic points.
 $\therefore$ There are only finitely many non-repelling periodic points of f .

Using this, we get that w is in the closure of repelling periodic points of f .

Since f is a polynomial, E contains all of $J(f)$ except for a finite number of points.

Since $J(f)$ has no isolated points, $J(f) \subseteq \bar{E}$.
This proves $J(f) \subseteq$ closure of repelling periodic pts. of f .

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